

AM115 In-class exercise: Sushi Restaurant

The owner of a sushi restaurant comes to you for advice. She wants to know how much fresh fish she should order each day. Fresh fish bought on day 1 can of course be used for sushi on that day and may also be refrigerated and used on day 2 but won't be up to snuff on day 3. The number of customers that come each day is a random number. The owner, from years of experience, knows roughly how the random number is distributed. Could you help her?

1. What kinds of questions would you ask her (then come up with the answers/guesses yourself ☺)?
2. Formulate the problem: what should be the state variable(s), what is the control, what is the reward, how to evolve the state variable(s)? After you have formulated the problem, we will provide some code to solve it.
3. Solve the optimal control problem using the `sushi_ddp.m` as the starting point. What is the optimal purchase policy if this is big chain with many stores?
4. What if this is a small business that's risk-averse and is content with making enough money to pay rent and other costs but will be under stress if that does not happen? In other words, how do we model risk? Modify `sushi_ddp.m` accordingly to find the optimal purchase policy in this case.
5. Modify the profit margin and other aspects of the problem to see how they affect the optimal purchase policy of the restaurant.